

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

NAME: NIVASINI VM

REG NO: 192511356

COURSE NAME: DESIGN AND ANALYSIS OF ALGORITHMS

COURSE CODE: CSA0613

COURSE OUTCOME COVERED

**CO 4 : Solve optimization problems using dynamic programming and greedy strategies.
(BL4, BL5)**

**Annexure A
Coding Challenge Task**

Q26. Egg Dropping Problem

A building has multiple floors, and a limited number of eggs are available to determine the highest floor from which an egg can be dropped without breaking. The goal is to minimize the number of attempts required in the worst case. Use Dynamic Programming to find the optimal strategy.

Input Format:

eggs = number of eggs

floors = number of floors

Sample Input:

eggs = 2

floors = 10

Expected Output:

Min Attempts = 4

AIM

To implement the Egg Dropping Problem using Dynamic Programming in Python and find the minimum number of attempts required in the worst case.

ALGORITHM

1. Start.
2. Read the number of eggs and floors.
3. Create a DP table for eggs and floors.
4. Initialize the base cases for 0 and 1 floor and 1 egg.
5. Try dropping an egg from every possible floor.
6. Calculate the worst-case attempts for each drop.
7. Store the minimum attempts in the DP table.
8. Display the minimum number of attempts.
9. Stop.

CODE

```
1 def egg_drop(eggs, floors):
2     dp = [[0] * (floors + 1) for _ in range(eggs + 1)]
3     for i in range(1, eggs + 1):
4         dp[i][0] = 0
5         dp[i][1] = 1
6     for j in range(1, floors + 1):
7         dp[1][j] = j
8     for i in range(2, eggs + 1):
9         for j in range(2, floors + 1):
10            dp[i][j] = float('inf')
11
12            for k in range(1, j + 1):
13                attempts = 1 + max(
14                    dp[i - 1][k - 1],
15                    dp[i][j - k]
16                )
17
18                dp[i][j] = min(dp[i][j], attempts)
19
20     return dp[eggs][floors]
21 eggs = int(input("Enter number of eggs: "))
22 floors = int(input("Enter number of floors: "))
23 print("Min Attempts =", egg_drop(eggs, floors))
```

OUTPUT

Output

```
Enter number of eggs: 2
Enter number of floors: 10
Min Attempts = 4
```

```
=== Code Execution Successful ===|
```

RESULT

Thus, the Egg Dropping Problem was successfully implemented using Dynamic Programming, and the minimum number of attempts for 2 eggs and 10 floors is 4.

ANALYSIS

The Egg Dropping Problem uses **Dynamic Programming** to avoid recalculating the same subproblems. The DP table stores the minimum number of attempts needed for a given number of eggs and floors. For every floor, the algorithm considers all possible dropping positions and selects the strategy with the minimum worst-case number of attempts.

TIME COMPLEXITY

$O(E \times F^2)$

Where:

- E = number of eggs
- F = number of floors

For each egg and floor combination, the algorithm checks every possible dropping floor.

SPACE COMPLEXITY

$O(E \times F)$

The DP table stores results for every combination of eggs and floors.